

Experiment 6.2

Determination of Capacitance with Multiple Dielectric Layers

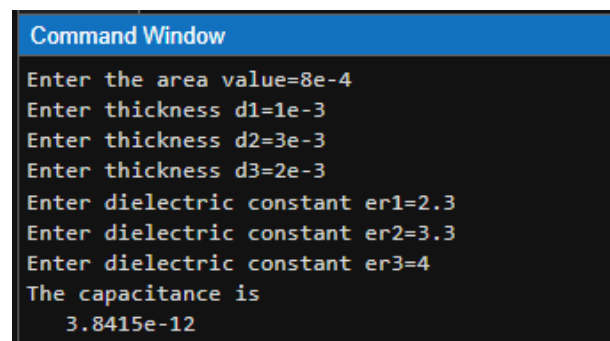
Code File Name: Exp6_2_MultipleDielectricsCapacitance.m

Code

```
clc;
A=input("Enter the area value=");
d1=input("Enter thickness d1=");
d2=input("Enter thickness d2=");
d3=input("Enter thickness d3=");
er1=input("Enter dielectric constant er1=");
er2=input("Enter dielectric constant er2=");
er3=input("Enter dielectric constant er3=");
epsilon0=8.854e-12;
C=epsilon0*A/((d1/er1)+(d2/er2)+(d3/er3));
disp('The capacitance is');
disp(C);
```

Command Prompt

```
Enter the area value=8e-4
Enter thickness d1=1e-3
Enter thickness d2=3e-3
Enter thickness d3=2e-3
Enter dielectric constant er1=2.3
Enter dielectric constant er2=3.3
Enter dielectric constant er3=4
The capacitance is
    3.8415e-12
```



The screenshot shows a MATLAB Command Window with a blue title bar labeled "Command Window". The window contains the same input and output sequence as the Command Prompt section above, with the final result of the capacitance calculation displayed as 3.8415e-12.

```
Command Window
Enter the area value=8e-4
Enter thickness d1=1e-3
Enter thickness d2=3e-3
Enter thickness d3=2e-3
Enter dielectric constant er1=2.3
Enter dielectric constant er2=3.3
Enter dielectric constant er3=4
The capacitance is
    3.8415e-12
```